

ASUS ROG Zephyrus Troubleshooting Report (Final Update)

1. Computer Identification:

- Make: ASUS
- Model: ROG Zephyrus
- Exact Model Number: GU502LV-B.537NB
- MFD (Manufactured Date): 2020-02-23
- Type: High-End Gaming Laptop
- CPU: Intel Core (exact generation/model not specified in conversation, but implied to be a gaming-grade Intel Core CPU based on ROG Zephyrus branding).

2. Initial Problem Description:

The laptop powers on, but its internal screen remains completely black, displaying no image. Furthermore, there is no display output whatsoever to an external monitor connected via HDMI.

3. Visual Evidence (from provided images and their relevance to the report):

- Image 1 (ASUS ROG Zephyrus - Keyboard Backlight): This image visually confirms the laptop is receiving power (illuminated keyboard and power button) but has no display output. The "ROG Zephyrus" branding is visible.
- Image 2 (ASUS ROG Zephyrus - Windows Login Screen with Time): This image documents a specific, transient event where the laptop successfully displayed the Windows 11 login screen, with the time displayed as 5:17, Monday, July 21. This photo is critical evidence of intermittent display capability.

4. Symptoms Observed and Chronological Events:

- Initial State (Prior to July 21): The laptop consistently powered on (fans audible and spinning normally, keyboard backlight illuminated, laptop chassis getting warm/hot), but produced no display output on both the internal screen and an external monitor. No audible POST (Power-On Self-Test) beep codes were emitted. The laptop was unresponsive to typical hard reset procedures (e.g., holding the power button down for an extended period) once it was in this initial problematic "on" state.
- Transient Event (Monday, July 21, 5:17 AM): At this specific time, during a restart, the laptop's screen temporarily displayed the Windows 11 login screen. However, the screen then froze and became unresponsive, preventing any clicks or interaction.
- Current State (After the Transient Event): Following the temporary display and subsequent restart, the laptop reverted to its original symptom: no display, just a black screen, identical to the initial problem reported.

5. Troubleshooting Methods Attempted (Chronological):

- Initial Troubleshooting (Dates not specified, prior to July 21):
 - Hard Reset: The user attempted a hard reset (disconnecting AC adapter and holding the power button for an extended period, 15-30 seconds, then 30-60 seconds). However, the user noted difficulty in making the laptop respond to the power button for a full hard reset once it was already in the problematic "on" state.

- External Monitor Test: An external monitor was connected via an HDMI cable. Result: No image appeared on the external monitor. This was a critical test as it helped differentiate between an internal display/cable issue and a core graphics problem.
- Brightness Check: The user confirmed that the screen brightness was not simply turned down.
- Peripheral Disconnect: All external peripherals were disconnected during troubleshooting to rule out conflicts.

6. Diagnosis and Likely Cause:

- Initial Diagnosis (Based on consistent black screen): The initial symptoms (power on, no display internal/external, no beep codes, getting hot) strongly indicated a hardware failure related to the graphics processing unit (GPU) or its associated video output circuitry on the motherboard.
- Updated Diagnosis (Considering the transient display): The observation that the screen could briefly display the Windows login screen (on July 21) before reverting to a black screen is crucial. This indicates that the GPU and display pathways are not entirely dead, but are in a highly unstable or intermittently failing state.
- This points towards a complex hardware issue, such as:
 - Intermittent GPU/VRAM Instability: The GPU or its dedicated video memory (VRAM) may be failing intermittently or becoming unstable under specific conditions (e.g., as drivers load or components heat up).
 - Motherboard Component Failure: A subtle fault on the motherboard affecting power delivery or data communication to the GPU.
 - Deep Driver Instability (Less Likely for Black Screen Reversion): While a corrupted graphics driver could cause freezes, the reversion to a complete black screen on restart strongly suggests a hardware component is preventing stable display output.

7. Interaction with Manufacturer/Retailer and Recommended Next Steps:

- ASUS Support Contact: The user contacted ASUS Support regarding the issue.
- Redirection to Best Buy: ASUS Support advised the user to contact Best Buy, as the laptop was purchased from them, and provided a contact who could assist.
- Next Action: Given the complex and intermittent nature of the display issue, professional diagnosis and potential repair/replacement are necessary. Following ASUS's recommendation to engage with Best Buy remains the primary next step.



